Almost everyone, regardless of age, values the ability to function as independently as possible during everyday life. Health-care consumers (patients and clients) typically seek out or are referred for physical therapy services because of physical impairments associated with movement disorders caused by injury, disease, or health-related conditions that interfere with their ability to perform or pursue any number of activities that are necessary or important to them. Physical therapy services may also be sought by individuals who have no impairment but who wish to improve their overall level of fitness or reduce the risk of injury or disease. An individually designed therapeutic exercise program is almost always a fundamental component of the physical therapy services provided. This stands to reason because the ultimate goal of a therapeutic exercise program is the achievement of an optimal level of symptom-free movement during basic to complex physical activities.

To develop and implement effective exercise interventions, a therapist must understand how the many forms of exercise affect tissues of the body and body systems and how those exercise-induced effects have an impact on key aspects of physical function. A therapist must also integrate and apply knowledge of anatomy, physiology, kinesiology, pathology, and the behavioral sciences across the continuum of patient/client management from the initial examination to discharge planning. To develop therapeutic exercise programs that culminate in positive and meaningful functional outcomes for patients and clients, a therapist must understand the relationship between physical function and disability and appreciate how the application of the process of disablement to patient/client management facilitates the provision of effective and efficient health-care services. Finally, a therapist, as a patient/client educator, must know and apply principles of motor learning and motor skill acquisition to exercise instruction and functional training. Therefore, the purpose of this chapter is to present an overview of the scope of therapeutic exercise interventions used by physical therapists. We discuss models of disablement and patient/client management as they relate to therapeutic exercise and explore strategies for teaching and progressing exercises and functional motor skills based on principles of motor learning.

ТЕРАПЕВТИЧЕСКИЕ УПРАЖНЕНИЯ: IMPACT ON PHYSICAL FUNCTION

Of the many procedures used by physical therapists in the continuum of care of patients and clients, therapeutic exercise takes its place as one of the key elements that lies at the center of programs designed to improve or restore an individual's function or to prevent dysfunction.2 Definition of Therapeutic Exercise Therapeutic exercise2 is the systematic, planned performance of bodily movements, postures, or physical activities intended to provide a patient/client with the means to Remediate or prevent impairments Improve, restore, or enhance physical function Prevent or reduce health-related risk factors Optimize overall health status, fitness, or sense of well-being

Balance. The ability to align body segments against gravity to maintain or move the body (center of mass) within the available base of support without falling; the ability to move the body in equilibrium with gravity via interaction of the sensory and motor systems.2,66,74,88,124,127,128 Cardiopulmonary fitness. The ability to perform lowintensity, repetitive, total body movements (walking, jogging, cycling, swimming) over an extended period of time1,81; a synonymous term is cardiopulmonary endurance.

Therapeutic exercise programs designed by physical therapists are individualized to the unique needs of each patient or client. A patient is an individual with impairments and functional limitations diagnosed by a physical therapist who is receiving physical therapy care to improve function and prevent disability.2 A client is an individual without diagnosed dysfunction who engages in physical therapy services to promote health and wellness and to prevent dysfunction.2 Because the focus of this textbook is on management of individuals with physical impairments and functional limitations, the authors have chosen to use the term "patient" rather than "client" or "patient/client" throughout this text. We believe that all individuals receiving physical therapy services must be active participants rather than passive recipients in the rehabilitation process to learn how to self-manage their health needs.

Coordination. The correct timing and sequencing of muscle firing combined with the appropriate intensity of muscular contraction leading to the effective initiation, guiding, and grading of movement. It is the basis of smooth, accurate, efficient movement and occurs at a conscious or automatic level.123,127 Flexibility. The ability to move freely, without restriction; used interchangeably with mobility.

Mobility. The ability of structures or segments of the body to move or be moved in order to allow the occurrence of range of motion (ROM) for functional activities (functional ROM).2,134 Passive mobility is dependent on soft tissue (contractile and noncontractile) extensibility; in addition, active mobility requires neuromuscular activation.

Aspects of Physical Function: Definition of Key Terms

The ability to function independently at home, in the workplace, within the community, or during leisure and recreational activities is contingent upon physical as well as psychological and social function. The multidimensional aspects of physical function encompass the diverse yet interrelated areas of performance that are depicted in Figure 1.1. These aspects of function are characterized by the following definitions.

Muscle performance. The capacity of muscle to produce tension and do physical work. Muscle performance encompasses strength, power, and muscular endurance.2 Neuromuscular control. Interaction of the sensory and motor systems that enables synergists, agonists and antagonists, as well as stabilizers and neutralizers to anticipate or 2 THERAPEUTIC EXERCISE: IMPACT ON PHYSICAL FUNCTION FIGURE 1.1 Interrelated aspects of physical function.

Stability. The ability of the neuromuscular system through synergistic muscle actions to hold a proximal or distal body segment in a stationary position or to control a stable base during superimposed movement.50,127,134 Joint stability is the maintenence of proper alignment of bony partners of a joint by means of passive and dynamic components.85 The systems of the body that control each of these aspects of physical function react, adapt, and develop in response to forces and

physical stresses (stress! force/ area) placed upon tisses that make up body systems.81,84 Gravity, for example, is a constant force that affects the musculoskeletal, neuromuscular, and circulatory systems. Additional forces, incurred during routine physical activities, help the body maintain a functional level of strength, cardiopulmonary fitness, and mobility. Imposed forces and physical stresses that are excessive can cause acute injuries, such as sprains and fractures, or chronic conditions, such as repetitive stress disorders.84 The absence of typical forces on the body can also cause degeneration, degradation, or deformity. For example, the absence of normal weight bearing associated with prolonged bed rest or immobilization weakens muscle and bone.8,20,69,84,108 Prolonged inactivity also leads to decreased efficiency of the circulatory and pulmonary systems.1 Impairment of any one or more of the body systems and subsequent impairment of any aspect of physical function, separately or jointly, can result in functional limitation and disability. Therapeutic exercise interventions involve the application of carefully graded physical stresses and forces that are imposed on impaired body systems, specific tissues, or individual structures in a controlled, progressive, safely executed manner to reduce physical impairments and improve function.

definition of therapeutic exercise to address the full scope of soft tissue stretching techniques.

Exercise Safety

Regardless of the type of therapeutic exercise interventions in a patient's exercise program, safety is a fundamental consideration in every aspect of the program whether the exercises are performed independently or under a therapist's supervision. Patient safety, of course, is paramount; nonetheless, the safety of the therapist must also be considered, particularly when the therapist is directly involved in the application of an exercise procedure or a manual therapy technique.

Many factors can influence a patient's safety during exercise. Prior to engaging in exercise, a patient's health history and current health status must be explored. A patient unaccustomed to physical exertion may be at risk for the occurrence of an adverse effect from exercise associated with a known or an undiagnosed health condition. Medications can adversely affect a patient's balance and coordination during exercise or cardiopulmonary response to exercise. Therefore, risk factors must be identified and weighed carefully before an exercise program is initiated. Medical clearance from a patient's physician may be indicated before beginning an exercise program.

Types of Therapeutic Exercise Intervention

Therapeutic exercise procedures embody a wide variety of activities, actions, and techniques. The techniques selected for an individualized therapeutic exercise program are based on a therapist's determination of the underlying cause or causes of a patient's impairments, functional limitations, or disability. The types of therapeutic exercise interventions presented in this textbook are listed in Box 1.1. Additional exercise interventions are used by therapists for patients with neuromuscular or developmental conditions.2 NOTE: Although joint mobilization techniques are often classified as manual therapy procedures, not therapeutic exercise,2 the authors of this textbook have chosen to include joint mobilization procedures under the broad CHAPTER 1 Therapeutic Exercise: Foundational Concepts 3 BOX 1.1 Therapeutic Exercise Interventions • Aerobic conditioning and reconditioning • Muscle performance exercises: strength, power, and endurance training • Stretching techniques including muscle-lengthening procedures and joint mobilization techniques • Neuromuscular control, inhibition, and facilitation techniques and posture awareness training • Postural control, body mechanics, and stabilization exercises • Balance exercises and agility training • Relaxation exercises • Breathing exercises and ventilatory muscle training • Task-specific functional training fatigue, the relationship of fatigue to the risk of injury, and the importance of rest for recovery during and after an exercise routine. When a patient is being directly supervised in a clinical or home setting while learning an exercise program, the therapist can control these variables. However, when a patient is carrying out an exercise program independently at home or at a community fitness facility, patient safety is enhanced and the risk of injury or re-injury is minimized by effective exercise instruction and patient education. Suggestions for effective exercise instruction and patient education are discussed in a later section of this chapter.

The environment in which exercises are performed also affects patient safety. Adequate space and a proper support surface for exercise are necessary prerequisites for patient safety. If exercise equipment is used in the clinical setting or at home, to ensure patient safety the equipment must be well maintained and in good working condition, must fit the patient, and must be applied and used properly.

Specific to each exercise in a program, the accuracy with which a patient performs an exercise affects safety, including proper posture or alignment of the body, execution of the correct movement patterns, and performing each exercise with the appropriate intensity, speed, and duration. A patient must be informed of the signs of respond to proprioceptive and kinesthetic information and, subsequently, to work in correct sequence to create coordinated movement.72 Postural control, postural stability, and equilibrium. Used interchangeably with static or dynamic balance.[50,124,127]

Inherent in the integration and application of knowledge of the disablement process in health-care delivery is an understanding that the process is not unidirectional; that is, it is not necessarily unpreventable or irreversible.[14] Furthermore, it is assumed that in most instances, depending on factors such as the severity and duration of the pathological condition, a patient's access to quality health care as well as the motivation and desires of the patient, the progression of the process can indeed be altered and the patient's function improved.[2,14,132,133] An understanding and application of the disablement process shifts the focus of patient management from strict treatment of a disease or injury to treatment of the impact that a disease, injury, or disorder has on a patient's function as well as the identification of the underlying causes of the patient's dysfunction. This perspective puts the person, not solely the disease or disorder, at the center of efforts to prevent or halt the progression of disablement by employing interventions that improve a patient's functional abilities while simultaneously reducing or eliminating the causes of disability.[39,132,133] Models of Disablement

As mentioned, therapist safety is also a consideration to avoid work-related injury. For example, when a therapist is using manual resistance during an exercise designed to improve a patient's strength or is applying a stretch force manually to improve a patient's range of motion, the therapist must incorporate principles of proper body mechanics and joint protection into these manual techniques to minimize his or her own risk of injury.

Throughout each of the chapters of this textbook, precautions, contraindications, and safety considerations are addressed for the management of specific pathologies, impairments, and functional limitations and for the use and progression of specific therapeutic exercise interventions.

ПРОЦЕСС И МОДЕЛИ ИНВАЛИДНОСТИ

Several models that depict the process of disablement have been proposed over the past 40 years. The first two schema developed were the Nagi model[86,87] and the International Classification of Impairments, Disabilities, and Handicaps (ICIDH) model for the World Health Organization (WHO).[51] The ICIDH model was revised after its original publication, with adjustments made in the descriptions of the classification criteria of the model based on input from health-care practitioners as they became familiar with the original model.[47] The National Center for Medical Rehabilitation Research (NCMRR) integrated components of the Nagi model with the original ICIDH model to develop its own model.[89] The NCMRR model added interactions of individual risk factors, including physical and social factors, to the disablement process.

It has been said that the physical therapy profession is defined by a body of knowledge and clinical applications that are directed toward the elimination or resolution of disability.[110] Understanding the disabling consequences of disease, injury, and abnormalities of development and how the risk of potential disability can be reduced, therefore, must be fundamental to the provision of effective care and services, which are geared to the restoration of meaningful function for patients and their families, significant others, and caregivers.

The Disablement Process

Definition Disablement is a term that refers to the impact(s) and functional consequences of acute or chronic conditions, such as disease, injury, and congenital or developmental abnormalities, on specific body systems that compromise basic human performance and an individual's ability to meet necessary,

customary, expected, and desired societal functions and roles.60,86,143 Physical therapists most commonly provide care and services to people with physical disability. Social, emotional, and cognitive disablement can affect physical function and vice versa and, therefore, should not be disregarded or dismissed.58,60 Implications in Health Care Knowledge of the process of disablement provides a foundation for health-care professionals to develop an appreciation of the complex relationships among function, disability, and health.62,132 This knowledge, in turn, provides a theoretical framework upon which practice can be organized and research can be based, thus facilitating effective management and care of patients that is reflected by meaningful functional outcomes.40,62

Although each of these models uses slightly different terminology, each reflects a spectrum of disablement. Several sources in the literature have discussed or compared and contrasted the terminology and descriptors used in these and other models.2,39,51,59,60,86,87,89 Despite the variations in these models, each taxonomy reflects the complex interrelationships among the following.

Acute or chronic pathology

Impairments Functional limitations Disabilities, handicaps, or societal limitations The conceptual frameworks of the Nagi, ICIDH, and NCMRR models of disablement, although applied widely 4 PROCESS AND MODELS OF DISABLEMENT Use of Disablement Models and Classifications in Physical Therapy

 TABLE 1.1 Comparison of Terminology of Three Disablement Models Model Tissue/Cellular Level Organ/System Level Personal Level Societal Level

Nagi Active pathology Impairment Functional limitation Disability ICIDH* Disease Impairment Disability Handicap ICF † Impairment of body Activity limitation Participation restriction structure/function *International Classification of Impairments, Disabilities, and Handicaps. †International Classification of Functioning, Disability, and Health.

During the early 1990s physical therapists began to explore the potential use of disablement models and suggested that disablement schema and related terminology provided an appropriate framework for clinical decision making in practice and research.40,59,120 In addition, practitioners and researchers suggested that consistent use of disablementrelated language could be a mechanism to standardize terminology for documentation and communication in the clinical and research settings.45 The American Physical Therapy Association (APTA) subsequently incorporated an extension of the Nagi disablement model and related terminology into its evolving consensus document, the Guide to Physical Therapist Practice2 (often called the Guide), which was developed to reflect “best practice” from the initial examination to the outcomes of intervention. The Guide also uses the concept of disablement as a framework for organizing and prioritizing clinical decisions made during the continuum of physical therapy care and services.

Impairment of body structure (anatomical) and function (physiological) Activity limitation Participation restriction Impact of contextual factors (environmental and personal) on functioning, disability, and health The components of the ICF model are compared in Table 1.1 to the Nagi and ICIDH models of disablement.

To be consistent with the language of the Guide, Figure 1.2, which was configured for this textbook, depicts a model of disablement and the potential impacts of therapeutic exercise interventions on the disablement process. The impact of risk factors has also been included in this depiction of the process. Incorporating risk factors into the in clinical practice and research in many health-care professions, have been criticized for their perceived focus on disease and a medical-biological view of disability as well as their lack of attention to the person with a disability.21 In response to these criticisms, the WHO undertook a broad revision of its conceptual framework and system for classifying disability described in its ICIDH model. Through a comprehensive consensus process over a number of years, the WHO developed the International Classification of Functioning, Disability, and Health (ICF).52,131-133 This new conceptual model integrates functioning and disability and is characterized as a bio-psycho-social model of disablement that provides a coherent perspective of various aspects of health. The revised model was also designed to place less emphasis on disease and greater emphasis on how people affected by health conditions live.21,52,132,133 The ICF model consists of the following components of health and health-related influences.

FIGURE 1.2 Impact of therapeutic exercise on the disablement process.

model underscores the assumption that disability can be prevented, eliminated, or reduced if the risk of occurrence or severity of pathology, impairment, or functional limitation is reduced. The model also shows that effective interventions, in particular therapeutic exercise, can have a positive impact on every aspect of the disablement process.

NOTE: Shortly after the second edition of the Guide was published, the WHO adopted and disseminated the ICF model of disablement, with its newly developed conceptual framework and system for classifying functioning and disability.[52] Consequently, information about this model and its revised concepts and terminology were not incorporated into the Guide, and so the elements of the ICF model are not yet widely used by the physical therapy community. Because physical therapists need to be aware of the changing concepts and language of disablement, it is likely that the next edition of the Guide will reflect this information.

By choosing to use a model of disablement as part of the theoretical framework of practice, physical therapists have a responsibility to provide evidence that there are, indeed, links among the elements of the disablement process that can be identified by physical therapy tests and measures. It is also the responsibility of the profession to demonstrate that not only can physical impairments be reduced but functional abilities can be significantly enhanced by physical therapy interventions. This body of evidence has just begun to emerge during the past decade. Examples of some of this evidence are integrated into this chapter and interspersed throughout the textbook.

An overview of the key components of the process of disablement, using language consistent with that in the Guide, is presented in the following sections of this chapter, with additional discussion of risk factors and their potential impact on disability. The relationship of the disablement model to patient

Physical therapists in all areas of practice treat patients with a multitude of pathologies. Knowledge of these pathologies (medical diagnoses) is important background information, but it does not tell the therapist how to assess and treat a patient's dysfunction that arises from the pathological condition. Despite an accurate medical diagnosis and a therapist's thorough knowledge of specific pathologies, the experienced therapist knows full well that two patients with the same medical diagnosis, such as rheumatoid arthritis, and the same extent of joint destruction (confirmed radiologically) may have very different severities of impairment and functional limitation and, consequently, very different degrees of disability. This emphasizes the need for physical therapists to always pay close attention to the impact(s) of a particular pathology on function when designing meaningful management strategies to improve functional abilities.

Impairments are the consequences of pathological conditions; that is, they are the signs and symptoms that reflect abnormalities at the body system, organ, or tissue level.[2,36,58] Types of Impairment Impairments can be categorized as arising from anatomical, physiological, or psychological alterations as well as losses or abnormalities of structure or function of a body system. Physical therapists typically provide care and services to patients with impairments that affect the following systems.

Musculoskeletal Neuromuscular Cardiovascular/pulmonary Integumentary Most impairments of these body systems primarily are the result of abnormalities of physiological function or anatomical structure. Some representative examples of physical impairments commonly identified by physical therapists and managed with therapeutic exercise interventions are noted in Box 1.2.

Impairments may arise directly from the pathology (direct/primary impairments) or may be the result of preexisting impairments management and physical therapy interventions, specifically therapeutic exercise, is also discussed.

Pathology/Pathophysiology This first major component of the disablement model refers to disruptions of the body's homeostasis as the result of acute or chronic diseases, disorders, or conditions characterized by a set of abnormal findings (clusters of signs and symptoms) that are indicative of alterations or interruptions of structure or function of the body primarily identified at the cellular level.2,36 Identification and classification of these abnormalities of anatomical, physiological, or psychological structure or process generally trigger medical intervention based on a medical diagnosis.

(indirect/secondary impairments). A patient, for example, who has been referred to physical therapy with a medical diagnosis of impingement syndrome or tendinitis of the rotator cuff (pathology) may exhibit primary impairments, such as pain, limited ROM of the shoulder, and weakness of specific shoulder girdle and glenohumeral musculature during the physical therapy examination (Fig. 1.3 A&B). The patient may subsequently develop secondary postural asymmetry because of altered use of the upper extremity.

FIGURE 1.3 (A) Impingement syndrome of the shoulder and associated tendinitis of the rotator cuff (pathology) leading to (B) limited range of shoulder elevation (an impairment) are identified during the examination.

Furthermore, when an impairment is the result of multiple underlying causes and arises from a combination of primary or secondary impairments, the term composite impairment is sometimes used.120 For example, a patient who sustained a severe inversion sprain of the ankle resulting in a tear of the talofibular ligament and whose ankle was immobilized for several weeks is likely to exhibit a balance impairment of the involved lower extremity after the immobilization order is removed. This composite impairment could be the result of chronic ligamentous laxity and impaired ankle proprioception

CHAPTER 1 Therapeutic Exercise: Foundational Concepts 7 BOX 1.2 Common Physical Impairments Managed with Therapeutic Exercise Musculoskeletal • Pain • Muscle weakness/reduced torque production • Decreased muscular endurance • Limited range of motion due to • Restriction of the joint capsule • Restriction of periarticular connective tissue • Decreased muscle length • Joint hypermobility • Faulty posture • Muscle length/strength imbalances

Neuromuscular • Pain • Impaired balance, postural stability, or control • Incoordination, faulty timing • Delayed motor development • Abnormal tone (hypotonia, hypertonia, dystonia) • Ineffective/inefficient functional movement strategies Cardiovascular/Pulmonary • Decreased aerobic capacity (cardiopulmonary endu- rance) • Impaired circulation (lymphatic, venous, arterial) • Pain with sustained physical activity (intermittent claudication) Integumentary • Skin hypomobility (e.g., immobile or adherent scarring) A B

Regardless of the types of impairment exhibited by a patient, a therapist must keep in mind that impairments manifest differently from one patient to another. In addition, not all impairments are necessarily linked to functional limitations or disability. An important key to effective management of a patient's problems is to recognize functionally relevant impairments, in other words, impairments that directly contribute to current or future functional limitations and disability. Impairments that can predispose a patient to secondary pathologies or impairments must also be identified.

from the injury or muscle weakness due to immobilization and disuse.

Equally crucial for the effective management of a patient's dysfunction is the need to analyze and determine, or at least infer and certainly not ignore, the underlying causes of the identified physical impairments, particularly those related to impaired movement.[117,118] For example, are biomechanical abnormalities of soft tissues the source of restricted ROM? If so, which soft tissues are restricted, and why are they restricted? This information assists the therapist in the selection of appropriate, effective therapeutic interventions that target the underlying causes of the impairments, the impairments themselves, and the resulting functional limitations.

NOTE: The term now used by the WHO to denote functional limitation is "ability limitation," as defined in the ICF model of functioning and disability[21,52,131-133] (see Table 1.1).

Although most physical therapy interventions, including therapeutic exercise, are designed to correct or reduce physical impairments, such as decreased ROM or